Indian Institute of Science Education and Research Kolkata

Department of Physical Sciences

PH1101 :: Mechanics I
End-semester Examination (Autumn 2023)
Marks: 50, Duration: 2 hours 30 minutes

1. Using the definition of derivative, compute $\frac{d}{dx}\left(\frac{e^{2x}}{2}\right)$. [3]

2. Draw two curves $f_1(x) = 2x$ and $f_2(x) = x^2$ in the domain $[-2, 2]$. Shade the region between these two curves in the given domain and then find the area of the shaded region. [2+4]

3. Using the inner product $(u, v) = \frac{1}{\pi}\int_{-\pi}^{\pi} uv\, dx$, compute the squared norms of $\sin mx$ and $\cos mx$ where m is an integer. Are $\cos mx$ and $\cos nx$ orthogonal to each other if both m and n are half-integers such that $m \neq n$? [6+3]

4. Show that the total linear momentum of an isolated system of N interacting particles is a conserved quantity. [6]

5. Consider a system of two particles having mass m and $2m$. Find the position of the center of mass if positions of the particles are given by $\vec{r}_1 = \cos(\omega t)\hat{i} + \sin(\omega t)\hat{j}$ and $\vec{r}_2 = \cos(\omega t)\hat{i} - \sin(\omega t)\hat{j}$ respectively. [4]

6. Solve for the position $x(t)$ of a particle whose motion is described by the equation $\ddot{x}(t) = -16x(t)$. The particle is at rest at $t = 0$ and its position is at $x = 12$. What would be the position of the particle at $t = 2$? [4+3]

7. An observer at a sea beach near the Earth's equator finds that the Sun takes about 2 min and 24 seconds to fully disappear below the horizon during sunset. Estimate the radius of the Sun, in the unit of radius of the Earth r_E, if the distance to the Sun is $23400 r_E$. [5]

8. Show that the orbit of planet Jupiter of mass M moving around the Sun having mass $M_\odot$ is an ellipse. In polar coordinates, acceleration vector is given to be $\vec{a} = (\ddot{r} - r\dot{\theta}^2)\,\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\,\hat{\theta}$. [10]

G. M. Hossain

PART A: Answer any **one** question

A-1. Consider a simple harmonic oscillator that has two successive stops (zero speed), at times t_1 and t_2 with $t_2 > t_1$. If the maximum speed of the oscillator is u, then what is the amplitude of the oscillation? [2

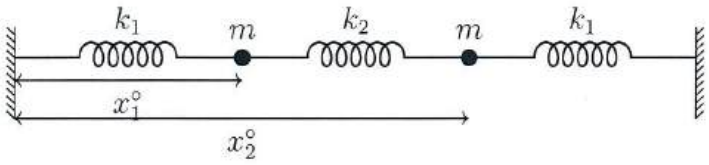
A-2. Consider the superposition of the two traveling waves $\sqrt{3}\cos(kx - \omega t)$ and $\sin(kx - \omega t)$. Plot the resulting waveform (plot it as a function of x) at $t = 0$. [2

PART B: Answer any **six** questions

B-1. Consider the following spring-mass system consisting of three springs and two masses. Each spring has zero unstretched length. [8

(a) Find x_1° and x_2°, the equilibrium position of the two masses. [4]

(b) For small displacement about these equilibrium positions, find the normal modes and corresponding frequencies. [4]



B-2. A massless spring with no mass attached to it hangs from the ceiling. Its length is L. A mass M is now hung on the lower end of the spring. Support the mass with your hand so the spring remains relaxed, then suddenly remove your supporting hand. The mass and spring will now oscillate. The lowest position of the mass during the oscillations is H cm below the place it was resting when you supported it. [8

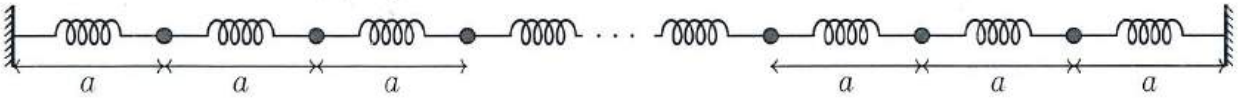
(a) Find the equation of motion. [2]

(b) Where is the equilibrium position of the mass? [1]

(c) What is the frequency of oscillation? What is the amplitude of the oscillation? [3]

(d) What is the velocity when the mass is $H/4$ cm below its original resting place? [2]

B-3. Consider a beaded string of N beads each of mass m (approximated as a long chain of the spring-mass system as shown in the figure). The beads are uniformly placed on the string, and the string has a uniform tension T. The horizontal distance between any two beads in equilibrium is a. The unstretched lengths of the springs are negligible. [8



(a) Find the equation of the motion of nth bead for the longitudinal mode of vibration. [2]

(b) Assuming normal mode vibration, find the normal mode frequency ω_m for m^{th} mode. [2]

(c) Plot the dispersion relation ω versus k. [2]

(d) Check whether we have $\omega_{N+2} = \omega_N$? [2]

B-4. A mass m is subject to a resistive force $-m\alpha v$ but no spring-like restoring force.

(a) Find the displacement x as a function of time in the absence of any driving force. [2]

Sangeet Bhattacharya

(b) At $t = 0$, a driving force $F = F_\circ \cos\omega t$ is switched on. Find the values of A and δ in the steady-state solution $x = A\cos(\omega t - \delta)$. Hint: you may use the following: [3]

$$A_{\text{ab}} = \frac{F_\circ}{m}\frac{\gamma\omega}{(\omega_\circ^2 - \omega^2)^2 + (\gamma\omega)^2}, \quad A_{\text{dis}} = \frac{F_\circ}{m}\frac{\omega_\circ^2 - \omega^2}{(\omega_\circ^2 - \omega^2)^2 + (\gamma\omega)^2},$$

where, the symbols carry their usual meanings.

(c) Write down the general solution (the sum of parts (a) and (b)) using the conditions that $x = 0$ and $dx/dt = 0$ at $t = O$. Sketch x as a function of t. [3]

B-5. Consider a Young's double slit interferometer and a monochromatic source with wavelength λ. d is the distance between the slits. Add another two slits, to create a linear four-slit interferometer with d distance between any two adjacent slits. Assume the slits to be thin. [8

(a) Calculate the amplitude of the waves from the four slits as they arrive at a place on the screen D distance away (as it happens in a double-slit experiment). [4]

(b) Calculate the resulting intensity pattern as a function of distance x on the screen from the central maximum or central minimum. Use small angle approximations, i.e., $\sin\theta \sim \tan\theta \sim \theta$. [2]

(c) Plot the intensity as a function of x. Compare it with the interference pattern of the double-slit case by plotting it on the same graph. [2]

B-6. Consider a Young's double slit interferometer and a *linearly-polarized* monochromatic source with wavelength λ. d is the distance between the slits. Assume the slits to be thin. A linear polarizer is placed in front of one of the slits. The pass axis of this polarizer is rotated by an angle ϕ *w.r.t.* the polarization of the original beam. [8

(a) Calculate the amplitude of the waves from the two slits as they arrive at a place on the screen D distance away. For simplicity, only consider the waves from the slits having the same polarization. [4]

(b) Calculate the resulting intensity pattern as a function of distance x on the screen from the central maximum. Use small angle approximations. [2]

(c) Plot the intensity as a function of x for $\phi = \pi/3$. [2]

B-7. (a) Suppose $Ae^{iax} + Be^{ibx} = Ce^{icx}$ for some nonzero constants A, B, C, a, b, c, and for all x. Prove that $a = b = c$ and $A + B = C$. [2+2]

(b) For a normal incidence, calculate the exact reflection and transmission coefficients without assuming $\mu_1 = \mu_2 \approx \mu_\circ$. You should start from Fresnel's equations (for *p-type*) and take appropriate limits to get normal incidence. Confirm that $I_R + I_T = I_\circ$. [2+2]

Sayan Bhattacharya

PH2102: End-semester examination (2022)

Time: 2.5 hours **Full marks: 50**

Answer all questions *Read both sides of this sheet*

Instructions:

- *This is a closed-books, closed-notes examination. The use of calculators is permitted.*
- **Throughout the question paper, c stands for the speed of light in free space.**

1. (a) A body of rest mass m_0 is moving with uniform speed v with respect to an inertial frame. If the total energy of the body in that frame is $2m_0c^2$, find v in units of c.

   (b) Consider a beta decay event $n \rightarrow pe^-\overline{\nu}$, where a neutron decays into a proton, an electron and an antineutrino.
   Take the rest masses of the neutron and the electron to be 939.6 MeV/c^2 and $0.5MeV/c^2$ respectively, and assume the rest mass of the antineutrino to be zero.
   If the total kinetic energy of the proton, the electron and the antineutrino is $0.8MeV$, what is the rest mass of the proton in units of MeV/c^2 ?

   [4 + 4]

2. Two spaceships approach each other along a line parallel to the surface of the earth, which for this problem can be taken to be plane and static. The two spaceships move with the same uniform speed in opposite directions, as measured by a static observer on the surface of the earth. The relative velocity of one spaceship with respect to the other is $0.7c$. Determine the speed of each spaceship, as measured by the static observer.

   [9]

3. Electrons in a projection television set are accelerated by a potential difference of $50keV$. Assume that the rest mass of the elrctron is $511keV/c^2$.

   (a) Calculate the speed of each electron in units of c, using the non-relativistic expression for kinetic energy, assuming that the electrons start from rest.

   (b) Now consider the electrons to be relativistic. Again, calculate their speed in the same units, and obtain the percentage difference between the two results.

   [3 + 5]

4. Consider a system consisting of electric charge as well as current distribution.

(a) Obtain the equation connecting the charge density ρ and the current density $\vec{j}$ at any point $(t, \vec{x})$, using the conservation of electric charge.

(b) From above, obtain a four-vector in terms of ρ and $\vec{j}$.

[3 + 5]

5. Consider two four-vectors A^μ, B^μ in Minkowski space, with components $A^\mu = (A_0, A_1, A_2, A_3)$ and $B^\mu = (B_0, B_1, B_2, B_3)$.

Suppose $A_0^2 - A_1^2 - A_2^2 - A_3^2 > 0$, and $B_0^2 - B_1^2 - B_2^2 - B_3^2 > 0$.

(a) Are the vectors A and B spacelike, null or timelike?

(b) Write down the expression for the inner product of A and B in terms of their components.

(c) Is it possible to go to some Lorenz frame where $A_0 \leq 0$? Give reasons.

(d) Find out whether the inner product of A and B can be zero.

[1 + 2 + 2 + 4]

6. A rigid pendulum having a massless rope of length l tied to a bob of mass m is undergoing oscillation in a frame.

(a) Write down the constraint equation(s) involving the co-ordinates of the of the bob, and thus identify the appropriate generalised co-ordinates.

(b) Obtain the equilibrium position of the bob, using the principle of virtual work.

(c) Find the Lagrangian of the bob in terms of the generalised co-ordinates, taking the point of suspension as the zero level of potential energy.

[3 + 3 + 2]

IISER–Kolkata
Classical Mechanics (PH3101)
End-semester examination

28 November 2023, 10 AM to 12.30 PM

Answer question 1 and any five from the rest. Try to write all parts of a question in the same place so that no part can get missed during evaluation.

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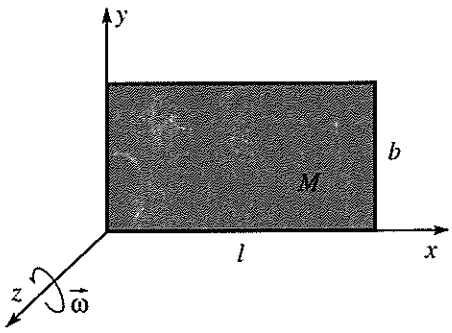
1. (a) For the heavy symmetrical top, what kind of motions do the derivatives of the three Euler angles $\dot{\phi}, \dot{\theta}, \dot{\psi}$ represent? [1+1+1]

   (b) Explain the terms 'limit cycle', 'quasiperiodicity' and 'chaos'. [1+1+1]

   (c) State the 'Least Action Principle'. Show that, for a one-dimensional system, the least action principle leads to the Lagrangian equation of motion. [1+3]

2. A rigid rectangular plate is rotating around the z-axis as shown in the figure.



   (a) Obtain the moment of inertia tensor of the body. [4]

   (b) Obtain the angular momentum vector. Explain your result briefly. [4]

3. Consider the linear system given by

$$\begin{aligned} \dot{x} &= -y \\ \dot{y} &= ax - 2y \end{aligned}$$

   where a is a parameter.

   (a) Obtain the general solutions of this system for $a = 0$, and $a = 2$. [3+3]

   (b) How will the character of the vector field change as a is varied continuously across 1? (schematically draw on your answerscript) [2]

4. (a) Write down the Hamilton-Jacobi equation for the simple harmonic oscillator and obtain the Hamilton's characteristic function. [4]

   (b) Is the following transformation canonical? [4]

$$Q = \log\left(\frac{1}{q}\sin p\right), \qquad P = q\cot p$$

5. (a) Suppose, at time $t = 0$, the dynamical state of a conservative Hamiltonian system is (q_0, p_0), and after a small interval δt, the state is (q_1, p_1). Express (q_1, p_1) as a function of (q_0, p_0) and show that the determinant of the Jacobian matrix of this function, in first order approximation, is unity. What does this result physically imply? [4]

   (b) For a one-dimensional system with the Hamiltonian

$$H = \frac{p^2}{2} - \frac{1}{2q^2}$$

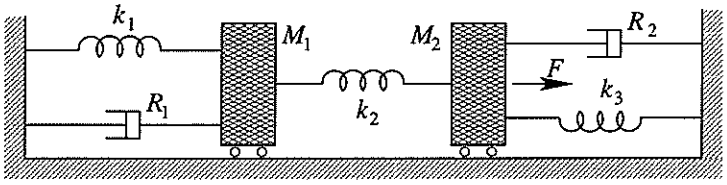
   Is the quantity $\left(\frac{pq}{2} - Ht\right)$ a constant of motion? [4]

6. A nonlinear system is given by the equations

$$\begin{aligned} \dot{x} &= -x - y \\ \dot{y} &= -x + x^2 \end{aligned}$$

   Schematically sketch the vector field in the phase space. [8]

7. Obtain the Lagrangian equations for the system shown below.



   Assume that the springs are relaxed in the equilibrium position. The force $F = A\sin\omega t$ is applied only on the mass M_2. [8]

Siddhartha Lal
02/12/23

PH3102 Quantum Mechanics **End-semester Examination**

Instructor: Dr. Siddhartha Lal **Autumn Semester, 2023**

Instructions: Answer all questions for full marks. Put your name and roll no. on both Question paper as well as the Answer script, and hand in both of them.

Date: 2nd December, 2023 **Full marks: 50** **Time: 2.5 hours**

1 SHO in the Heisenberg picture. [7 marks]

A particle is placed in the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ at time $t = 0$, where $|n\rangle$ is the nth eigenstate of the 1D simple harmonic oscillator $H = p^2/2m + m\omega^2 x^2/2$.
(a) Obtain compact expressions for the operator $x(t)$ in the Heisenberg picture in terms of the operators $x(t = 0)$. [3 marks]
Hint: You may find the BCH identity useful:

$$e^X Y e^{-X} = Y + \frac{1}{1!}[X, Y] + \frac{1}{2!}[X, [X, Y]] + \ \ldots \tag{1}$$

(b) Now, calculate the expectation values of the operator $x^2(t)$ in the state $|\psi\rangle$ [4 marks].

2 Anisotropic two-spin model. [12 marks]

Two spins S_1 and S_2 are interacting with each other through the Hamiltonian

$$H = J_\perp \left(S_1^x S_2^x + S_1^y S_2^y\right) + J_\parallel S_1^z S_2^z \ . \tag{2}$$

The total spin operator is defined as $\vec{S}_{\text{tot}} = \vec{S}_1 + \vec{S}_2$.
(a) Calculate the commutators of the Hamiltonian with the operators $S^x_{\text{tot}}, S^y_{\text{tot}}$ and S^z_{tot}. **Hint:** Recall the commutation relation $[S_\alpha, S_\beta] = i\hbar\epsilon_{\alpha\beta\gamma}S_\gamma$, where $(\alpha, \beta, \gamma) \in (x, y, z)$. [8 marks]
(b) Using the results of part (a), compute the commutator of the Hamiltonian with $\vec{S}_{\text{tot}}$. What value does it take for the case $J_\perp = J_\parallel$? Which operator is conserved in time for this case, and which symmetry of the Hamiltonian does this correspond to? [4 marks]

3 Density matrix and the Liouville–von Neumann equation. [12 marks]

Consider an arbitrary state $|\psi\rangle = a\,|\uparrow\rangle + b\,|\downarrow\rangle$ for a spin-1/2 object, where the states $|\uparrow\rangle$ and $|\downarrow\rangle$ are the eigenstates of σ^z with eigenvalues 1/2 and −1/2 respectively. Such a state is placed in a magnetic field B, so that it evolves according to the Hamiltonian

$$H = B\sigma^z \ . \tag{3}$$

(a) Use the time evolution operator to find the time-evolved state $|\psi(t)\rangle$ at time t. [2 marks]
(b) Find the corresponding density matrix $\rho(t)$ at time t, where a density matrix is defined as $\rho = |\psi\rangle\langle\psi|$. [3 marks]
(c) Calculate the commutator $[H, \rho]$ between $\rho(t)$ and the Hamiltonian H. You can use the 2×2 matrix representation of ρ and H for this. [3 marks]
(d) By comparing the respective expressions, derive a relation between $d\rho/dt$ and $[H, \rho]$? This relation is called the Liouville–von Neumann equation. [Here, we consider the derivative of a

matrix as simply element-wise differentiation.] [2 marks]
(e) What happens to the commutator when $b = 0$? What does this mean for the density matrix according to the Liouville–von Neumann equation? [2 marks]

4 SHO in an electric field. [7 marks]

We promise that this is the last SHO problem of this course. Consider adding an electric field ε to the SHO Hamiltonian as a perturbation:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + \varepsilon x \ . \tag{4}$$

(a) Calculate the correction E_n^{low} to the energy eigenvalues $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$ upto second order in ε using perturbation theory. Recall that $E_n^{(1)} = \langle n|V|n\rangle$ and $E_n^{(2)} = \sum_{n' \neq n} \frac{|\langle n'|V|n\rangle|^2}{E_n^{(0)} - E_{n'}^{(0)}}$. [5 marks]
(b) Find the polarisability α resulting from this correction, using the definition [2 marks]

$$\alpha = -\frac{\mathrm{d}^2 E_n^{\text{low}}}{\mathrm{d}\varepsilon^2} \ . \tag{5}$$

5 Perturbing a two-dimensional box. [12 marks]

A particle of mass m is placed in a two-dimensional infinitely deep box potential of extent a in both the x and y directions. Further, a perturbation $V(x, y) = -Kxy$ is applied to this system. It is given that the eigenfunctions of a 1D infinite well in the x direction are of the form $\psi_n(x) = \sqrt{\frac{2}{a}}\sin(n\pi x/a)$ with eigenvalues $E_n = \hbar^2\pi^2 n^2/2ma^2$, for $n = 1, 2, \ldots$
(a) Find the eigenfunctions and eigenvalues for the ground state as well as the doubly degenerate first excited states of the unperturbed problem. [3 marks]
(b) Compute the first and second order correction to the ground state energy. [5 marks]
(c) Compute the first order correction to the energy of the first excited states. Also clearly write the appropriate linear combinations of the degenerate unperturbed states to which these corrections apply. [4 marks]
Hint: You may find the following integrals helpful:

$$\int_0^1 dz\ z\sin^2(m\pi z) = 0.25 \text{ for } m = 1, 2, \ldots; \quad \int_0^1 dz\ z\sin(\pi z)\sin(2\pi z) = -\frac{8}{9\pi^2} \tag{6}$$

Siddhartha Lal
02/12/23

IISER Kolkata
End Semester Examination, 2023
PH 3103: Mathematical Methods of Physics
3rd Year BSMS
Time: Two hours and thirty minutes
Attempt all questions.

1. a) Show that $u = x^3 - 3y^2x$ is a harmonic function. Find the corresponding conjugate function, and hence write down the analytic function $f(z)$ where $z = x + iy$. (4)
b) Evaluate the integral $\int_C \frac{\sin z}{z - \frac{\pi}{4}} dz$.
where C is the circle $|z| = 2$. (3)
c) Evaluate the integral
$\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta}$. (3)

2. a) Find the relevant Green's function for the equation $\frac{d^2y}{dx^2} + \omega^2 y = f(x)$ where $y(x=0) = y(x=l) = 0$. (6)
b) Using the generating function $g(x,t) = \exp[\frac{x}{2}(t - \frac{1}{t})]$ for the Bessel function $J_n(x)$, arrive at the relation
$\sin x = 2\Sigma_{n=0}^{n=\infty}(-1)^n J_{2n+1}(x)$. (4)

3. a) Show that if a matrix A and its transpose conjugate $(A^\dagger)$ are equal, the eigenvalues are real. Also, the eigenvectors corresponding to distinct eigenvalues are orthogonal. (3+3)
b) Show that the three vectors $A = \frac{1}{\sqrt{2}}(1, -i, 0), B = \frac{1}{\sqrt{2}}(1, i, 0), C = \frac{1}{\sqrt{2}}(0, 0, 1)$ form an orthonormal set. (4)

4. a) Find the Fourier transform of the functions (3+4)
(i) $f(x) = e^{-x^2}$
(ii) $f(x) = 1$ for $-l \leq x \leq +l$
$= 0$ for $|x| \geq l$.

(b) The Fourier transform of the function $f(x)$ is $g(k)$. Arrive at the Fourier transform of $\frac{df}{dx}$. (3)

5. a) Using the properties of the multiplication table, construct the group multiplication table for the group $G_6 = \{e, a, a^2, b, ba, ba^2\}$, with $ab = ba^2$. (3)

b) Consider the group $G = \{a, a^2, ...a^{m-1}, a^m = e\}$ with $O(G) = m$. Show that this is an abelian group. (3)

c) Let $G_1 = \{e_1, a_1, b_1, ...\}$ and $G_2 = \{e_2, a_2, b_2, ...\}$ be two groups and G is the direct product $G_1 \otimes G_2$. If $G_1' = \{(e_1, e_2), (a_1, e_2), (b_1, e_2), ...\}$, show that G_1' is an invariant subgroup of G. (4)

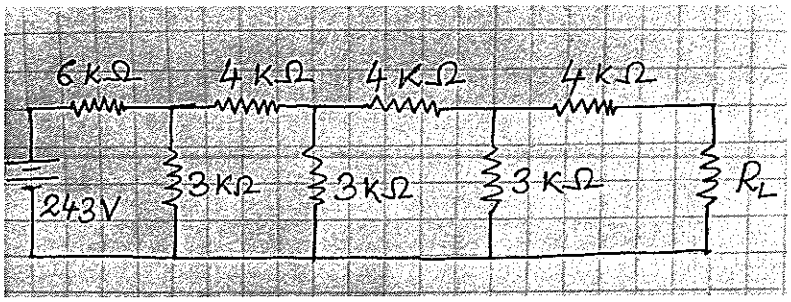
30/11/23

30/11/2023

End Semester Examination (Autumn 2024)
Course: PH3104 : Electrical Circuits and Electronics

Time: 2 hrs 30 mins | Total Marks: 50

1. (a) Calculate the load current in the following figure for the load resistance (R_L) of 3 kΩ: [3]



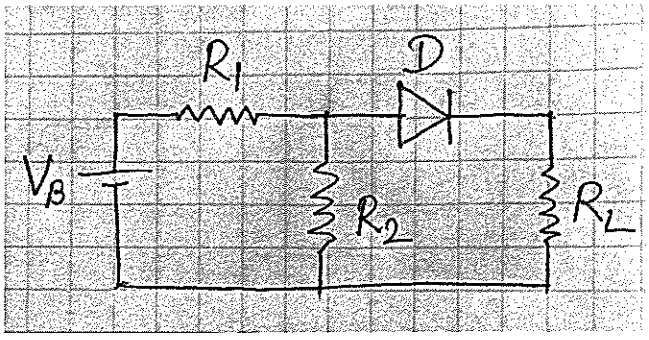
(b) Describe Half-voltage method for measuring R_{Th}. [2]

(c) Derive the expression of resonant frequency (f_r) and Bandwidth (BW) for a RLC parallel resonant circuit in terms of R, L and C. [2+3]

2. (a) The barrier potential developed across an open-circuited p-n junction aids the flow of minority carriers. Explain how the flow of charge is counter-balanced. [1]

(b) Draw the circuit diagram for a positive diode clipping circuit. Sketch the output waveform and discuss the observations. If the reference voltage is set to be zero and the Si diode is used in the circuit that is a non-ideal, then what will be the maximum positive and negative output voltages when an AC input voltage (Peak voltage of 50 V) is used? [1+(1+2)+1]

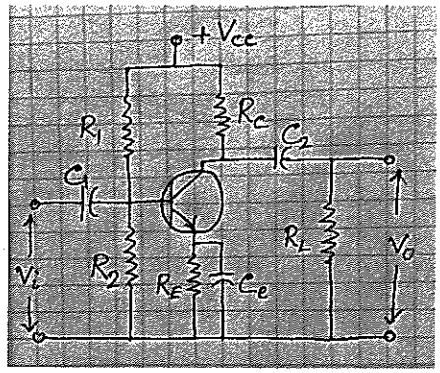
(c) Determine the current through the load resistor (R_L) in the following circuit, consider the Si diode used is a non-ideal diode. Also consider V_B = 15 V, R_1 = 6 kΩ, R_2 = 4 kΩ and R_L = 2 kΩ. [2]



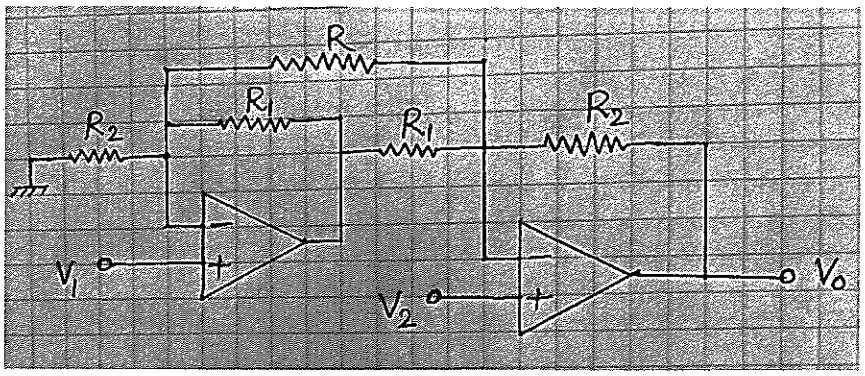
(d) Draw the circuit diagram for a AND logic gate and write down the truth table. [2]

3. (a) Define hybrid parameters for a transistor and draw the h-parameter equivalent circuit. Discuss how one can determine the h-parameters experimentally. [2+1+2]

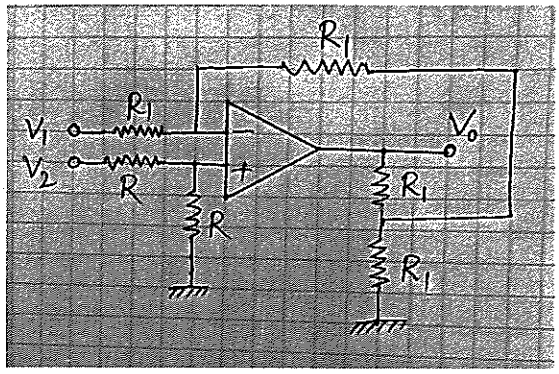
(b) In the CE amplifier circuit, R_1 = 72 kΩ, R_2 = 8 kΩ, R_L = 2 kΩ, R_E = 700 Ω and R_C = 2 kΩ and V_{CC} = 15 V. Assume β = 120 and V_{BE} = 0.7 V. Draw the dc load line and find the Q-point. [5]



4. (a) Discuss the differential amplifier with circuit diagram. [2]
   (b) The following circuit shows an instrumentation amplifier used as the input amplifier of many electronic instruments and to amplify the signals at the output of transducers for driving the indicator. Calculate the voltage gain ($A_V = \frac{V_o}{V_2 - V_1}$) of the amplifier. How can you vary the gain? [4+1]



   (c) Determine the output voltage V_o in the circuit below in terms of V_1 and V_2. [3]



5. (a) Give the logic symbol, Boolean expression and the truth table of a two-input NAND gate. [2]
   (b) Implement the Boolean function $Y = AB + \bar{A}C$ with NAND gates only. [2]
   (c) Simplify by Boolean algebra, $Y = \bar{A}B\bar{C} + \bar{A}C\bar{B} + A\bar{B}C + \bar{A}BC$ [2]
   (d) Simplify the Boolean expression $Y = \bar{A}BC + ABC + A\bar{B}C + AB\bar{C}$ using a Karnaugh map and draw the simplified logic circuit. [3 + 1]

PH4101: Basic Condensed Matter Physics Final Examination

Instructor: Dr. Kamaraju Natarajan

Department of Physical Sciences, IISER Kolkata

Total Marks 50 **Duration:** 2 hr 30 minutes

Name:

Roll No:

Important Instructions

There are three sections. Section-I contains 7 questions (1 mark each), section-II contains 4 questions (2 marks each) and section-III has 10 questions (5 marks each). In section-III, you have to answer any 7 questions. You should attempt all the sections. If in answering a question you sketch a diagram, please provide meaningful labels. The question paper should be attached along with your answer sheets.

Useful Identities

$\int_0^\infty \frac{dx}{e^x-1} = \infty$	$\int_{x_0}^\infty \frac{dx}{e^x-1} = -log(x_0-1)+x_0$
$\int_0^\infty \frac{xdx}{e^x-1} = \frac{\pi^2}{6}$	$\int_0^\infty \frac{x^3dx}{e^x-1} = \frac{\pi^4}{15}$
$\int_0^\infty \frac{x^2dx}{e^x-1} = 2\zeta(3) = 2(1+\frac{1}{2^3}+\frac{1}{3^3}+\frac{1}{4^3}+...)$	$\int_0^\infty \frac{x^4dx}{e^x-1} = 24\zeta(5) = 24(1+\frac{1}{2^5}+\frac{1}{3^5}+\frac{1}{4^5}+...)$
$\int_0^\infty dx\ e^{-\alpha x^2} = \frac{\sqrt{\pi}}{2\alpha^{1/2}}$	$\int_0^\infty dx\ x^2 e^{-\alpha x^2} = \frac{\sqrt{\pi}}{4\alpha^{3/2}}$
$\int_0^\infty dx\sqrt{x}\ e^{-x} = \frac{\sqrt{\pi}}{2}$	$\int_0^\infty dx\ x\ \sin(bx)e^{-ax} = \frac{\sqrt{2}ab}{(a^2+b^2)^2}$
$\cos(A-B) = \cos A\cos B + \sin A\sin B$	$\sin(A+B) = \sin A\cos B + \cos A\sin B$

Universal Constants	Conversion Factors
$m_e = 9.11\times10^{-31}kg$	$1eV = k_B.(11,605K)$
$c = 2.99999\times10^8 m/s$	$k_B = 8.617\times10^{-5}eV.K^{-1}$
$h = 6.626\times10^{-34}J.s$	$\hbar = 6.582\times10^{16}eV\cdot s$
$e = 1.6\times10^{-19}C$ or 4.80×10^{-10} esu	$hc = 1240eV\cdot nm$
$k_B = 1.381\times10^{-23}J/K$	1 Tesla $= 10^4$ Gauss
$\varepsilon_0 = 8.85\times10^{-12}Fm^{-1}(Fm^{-1} = C^2/(Nm^2))$	$1N = 10^5 Dynes$
Bohr Radius: $a_0 = \frac{4\pi\varepsilon_0\hbar^2}{e^2m_e} = 5.29\times10^{-11}m$	
Avogadro constant, $N_A, = 6.02214076\times10^{23}$/mol	

N. Kamaraju

Section - I (7 x 1 = 7)

Answer all the questions

1. The value of the angle between the tetrahedral bonds of diamond is (Hint: The angles between the tetrahedral bonds of diamond are same as the angle between the body diagonals of a cube.)

   A. $100°18'$ B. $60°$. C. $120°$ D. $109°28'$

2. Vanadiuum (V) occurs in BCC lattice (M=50.94 g/mole and density is 5.8 g/cm^3). The highest linear density of atoms (atoms/m) is

   A. The highest density would be along [100] and the density is 2.75 $\times 10^9$ atoms/m
   B. The highest density would be along [101] and the density is 3.75 $\times 10^9$ atoms/m
   C. The highest density would be along [111] and the density is 3.75 $\times 10^9$ atoms/m
   D The highest density would be along [110] and the density is 1.75 $\times 10^9$ atoms/m

3. An X-ray beam of wavelength $\lambda = 0.84A°$ is used to study the NaCl crystal. If the first peak of first order of Bragg reflection, is observed at an angle of $8°35'$, the following is true (if there is enough intensities in the higher order Bragg diffraction).

   A. the highest lattice plane spacing is $2.835A°$ and the maximum order of diffraction that can be detected is 4
   B. the highest lattice plane spacing is $2.035A°$ and the maximum order of diffraction that can be detected is 4
   C. the highest lattice plane spacing is $2.835A°$ and the maximum order of diffraction that can be detected is 2
   D. the highest lattice plane spacing is $2.835A°$ and the maximum order of diffraction that can be detected is 1

4. Drude got the Lorentz number (ratio of thermal conductivity to electrical conductivity) almost only half of the actual value since

   A. Only his calculation of velocity was over-estimated.
   B. His calculation of heat capacity was correct, but his velocity was over estimated.
   C. Only his calculation of velocity was correct.
   D. His calculation of velocity was under-estimated and heat capacity was correct.
   E. His calculation of heat capacity was over-estimated whereas the velocity was under-estimated.

5. Consider a longitudinal wave, $u_s = ucos(\omega t - sKa)$ propagating in a mono-atomic linear lattice of atoms of mass M, spacing a, and nearest neighbor coupling constant, C. The total energy of the wave is

   A. $E = (\frac{1}{2}) * M \sum_s (u_s)^2 + (\frac{1}{2}) * C \sum_s (u_s - u_{s+1})^2$ where s runs over all the atoms
   B. $E = (\frac{1}{2}) * M \sum_s (du_s/dt)^2 + (\frac{1}{2}) * C \sum_s (u_s - u_{s+1})^2$ where s runs over all the atoms
   C. $E = (\frac{1}{4}) * M \sum_s (u_s)^2 + (\frac{1}{4}) * C \sum_s (u_s - u_{s+1})^2$ where s runs over all the atoms
   D. $E = (\frac{1}{4}) * M \sum_s (du_s/dt)^2 + (\frac{1}{4}) * C \sum_s (u_s - u_{s+1})^2$ where s runs over all the atoms
   E. $E = M \sum_s (du_s/dt)^2 + C \sum_s (u_s - u_{s+1})^2$ where s runs over all the atoms

6. The following statement/statements denote the main consequences of the harmonic theory

   A. A single lattice wave does not decay or change form with time, i.e two lattice waves do not interact.
   B. adiabatic and isothermal elastic constants are never equal.
   C. the elastic constants are independent of pressure and temperature.
   D. there can be thermal expansion.

7. The following statement/statements is/are TRUE regarding electron-phonon scattering

   A. At T>> T_{Debye}, the resistivity is linearly proportional to T^2.

   B. At T>> T_{Debye}, the resistivity is linearly proportional to T^3.

   C. At T<< T_{Debye}, the resistivity is $\approx T^5$ in general for alkali metals.

   D. At T<< T_{Debye}, the electron scattering rate goes as $q_{ph}(1 - cos(\phi))$ where the q_{ph} is the phonon wavevector and ϕ is angle between the incoming electron and the scattering electron.

Section - II (4 x 2 = 8)

Answer all the questions

8. The kinetic energy of a three dimensional gas of N electrons at 0K is

   A. $\frac{2N\epsilon_F}{5}$ B. $\frac{N\epsilon_F}{5}$ C. $\frac{3N\epsilon_F}{5}$ D. $\frac{4N\epsilon_F}{5}$

9. The spring constant of NaCl single crystal is 11 N/m along [100] direction and at the gamma point. The wavelength of IR radiation that will be strongly reflected by this crystal is (Useful Information: M_{Na}=23 g/mol and M_{Cl}=35.45 g/mol, and you may also would like to think about how reflectivity and permittivity/refractive index change at LO/TO/and phonon resonance frequency)

   A. 61 μm which is the wavelength of the LO phonon. IR reflectivity is also minimum at LO phonon resonance.

   B. 91 μm which is the wavelength of the transverse acoustic phonon. IR reflectivity is also minimum here.

   C. 9 μm which is the wavelength of the longitudinal acoustic phonon. IR reflectivity is also maximum here.

   D. 61 μm which is the wavelength of the TO phonon. IR reflectivity is also maximum at TO phonon resonance.

10. The following statement/statements is/are FALSE

    A. In a simple square lattice in 2D, the kinetic energy of the electron at a corner of the first zone is higher than that of a free electron at the mid point of a side face of the zone by a factor 2.

    B. Bloch functions do not exist at the forbidden regions of energy bands.

    C. Mono-valent electron compounds are insulators as their fermi surface touches the first Brilluoin zone.

    D. The alkaline earth metals have two valence electrons per primitive cell; they could be insulators, but the bands overlap in energy to give metals, but not very good metals. Examples for Rare Earth metals are beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), and radium (Ra).

11. Assume that a new quasi 2D material (lattice constant is $2.25A^\circ$) is invented in a laboratory and you are asked to determine the carrier density in cm^{-2} of this 2D material using fermi surface measurement experiment like De Haas Van Alphen effect and De-Shubnikov oscillations in resistivity. For example, the resistivity and other material properties exhibit oscillations as the field varies. These oscillations are periodic in $\Delta(\frac{1}{B})$ and this periodicity is $4.5 \times 10^{-8} G^{-1}$.Calculate the carrier density in cm^{-2} of this 2D material
    (Hint: The energy dispersion of this 2D material is well described by tight binding approximation. Use CGS units. In CGS, electron charge is $4.80326 \times 10^{-10} esu$; c = 3×10^{10} cm/s; h = 6.626196×10^{-27} erg.s)

Section - III (7 x 5 = 35)

Answer any 7 questions

12. Say an X-ray with wavevector k is scattered at a volume dV of a sample and the outgoing X-ray has a wavevector k'. The local electron concentration at the sample is n(r). The scattering wave amplitude in this case is given as $F = \int dV n(r) e^{(i(\mathbf{k}-\mathbf{k}').\mathbf{r})}$
    (a) Now, for a crystal of N cells, write down the scattering amplitude (in terms of structure factor) when the diffraction condition is met. (1)
    (b) Assume there are 's' atoms in the unit cell and for this case, derive the most general form, of structure factor in terms of atomic form factor. (2)
    (c) Derive the classical expression of atomic form factor in spherical polar co-ordinates. (2)

13. For a 1D mono-atomic lattice, (a) Show that $k_1 = \frac{\pi}{a}$ and $k_2 = \frac{4\pi}{5a}$ describe the same elastic wave/phonon that satisfy the aliasing condition, i.e., the mathematical condition that makes these two waves indistinguishable. (1)
(b) Make a plot of $\cos(k_1x)$ and $\cos(k_2x)$ vs x/a on a 1D schematic montoatomic lattice. Prove the conclusion from (a) using these plots. Indicate the wavelength of these two vibration waves (1)
(c) For a 1D diatomic lattice with two types of atoms with masses, M_1 and M_2, sketch the dispersion relation for $M_1 \neq M_2$ and $M_1 = M_2$. (1.5)
(d) When $M_1 = M_2$, do you think the lattice mentioned in (c) will still have optical phonons. Explain your answer. (1.5)

14. (a) Assuming electron conductivity effective mass, $m^* = 0.26\ m_e$ for Si, where m_e is the electron rest mass in free space. Find the average kinetic energy (thermal energy) of conduction electrons in n type Si near thermal equilibrium at temperature T = 300 K. Find the thermal velocity of electrons.
(b) Given electron mobility $\mu = 280\ cm^2/(Vs)$ at a doping level of $10^{18}\ cm^{-3}$, find the average scattering time τ
(c) Many textbooks quote an electron mobility $\mu = 1500\ cm^2/(Vs)$ for Si, much higher than the above value. What could be the origin of this high mobility? What is the average scattering time τ corresponding to this high mobility value?
(d) These two electron mobility values and corresponding scattering times give you a sense of orders of magnitude of the quantities describing electron transport in Si. Calculate the mean free path using these two mobilities and comment on what should be dimensions of modern CMOS devices such as field effect transistors (FETs) using Si?
(e) Assuming ionized impurity and lattice vibration scatterings are the two predominant scattering mechanisms, and based on the two mobility values and corresponding τ values in (b) and (c), estimate the average ionized impurity scattering time τ_i and average phonon scattering time τ_{ph}. Here, τ_i and τ_{ph} should be interpreted as $< \tau_i >$ and $< \tau_{ph} >$, respectively. Compare these two scattering times to find which mechanism is more limiting on the mobility of electrons for n type Si doped to $10^{18}\ cm^{-3}$ at 300 K. (5)
(Hint: This problem can be solved from free electron theory of metals itself)

15. (a) When there is thermal vibration u(R), the periodicity of the crystal is perturbed slightly. Start with a most general expression for the periodic potential in a solid as a sum pf potentials summed over all Bravais lattice points. Now show for vibrating lattice, the total potential has a pertubation due to small vibration using Taylor series expansion.(1)
(b) Consider forward scattering of electrons from k_e to k_e^s with phonons of momentum $\hbar q_{ph}$ in an ideal metal. The angle between k_e and k_e^s, is θ. Derive relationship that exists between the electrical resistivity and temperature (i) at high temperatures, $T >> T_{Debye}$, and (ii) at very low temperatures, $T << T_{Debye}$. Show the scattering process with a schematic on the Fermi surface. (Hint: The electron scattering rate is proportional to $|k_e - k_e^s|$ weighed by 1-cos(θ)). (1+3)

16. Derive the propagation condition for (a) transverse and (b) longitudinal plasma wave oscillations of free electrons in a metal when an electromagnetic wave is shined. For simplicity, consider a isotropic metal medium where electromagnetic wave is shined along the z direction. (Hint: Write down the Maxwell equation in the given situation here and consider AC Drude model. Then derive the general propagation equation of both the modes in a metal and then derive the condition for transverse and longitudinal plasma wave oscillations of free electrons) (3+1+1)

17. Consider a simple model of cubic ionic crystals (polar crystals) with two atoms in a primitive unit cell, e.g NaCl, CsCl and ZnS. Also let the transverse oscillating displacements of +ve and -ve ions be represented by, u_+ and u_- respectively. Let there be N primitive cells occupying the volume of V.
(a) Now, if one shines an electromagnetic wave on this polar crystal perpendicular to the transverse oscillating charges, derive the equations of motion of this system of +ve and -ve ions of masses m_1 and m_2. Show that these two equations can be combined to write an effective equation of motion of a damped harmonic oscillator (Lorentz-Drude oscillation equation) by adding a phenomenological damping term, $\gamma\frac{dr}{dt}$ where $r = u_+ - u_-$
(b) The solution to the above Lorentz-Drude oscillation equation is $r = \frac{QE/\mu}{\omega_0^2 - \omega^2 - i\gamma\omega}$ where Q is the effective charge, μ is the reduced mass of the pair of ions and ω is the angular frequency of the electromagnetic wave that is shined and ω_0 is the resonant frequency of the transverse oscillations. Now, derive the expression for permittivity by considering semiconductors where only bound charges are considered. (Hint: Derive the electrical polarization in the present case and you

should be able to see the permittivity emerging).
(c) For the case when there is no damping ($\gamma = 0$), derive the relationship between the longitudinal frequency of oscillation and transverse frequency of oscillations. (2+2+1)

18. Considering a 1D periodic potential for electrons (mass, m), $U(x) = U(x+a)$ where a is the periodicity. Also, let the crystal be symmetric about x=0, and now
(a) derive the most general form of central wave equation for electrons $(\lambda_k - \varepsilon)C_k + \sum_G U_G C_{k-G} = 0$ where $\lambda_k = \frac{\hbar^2 k^2}{2m}$ and G is the reciprocal lattice vector. (b) Write down the central wave equation for electrons at the first Brilluoin zone boundary, k = $\pm\frac{1}{2}G$ with the assumption that only the Fourier coefficients, $U_G = U_{-G} = U$ exist in the Fourier expansion of U(x). (Hint: Consider Bloch wave function for the electrons) (4+1)

19. The first order tight binding approximation gives $\varepsilon_k = -\alpha - \gamma \sum_m exp(-iK.\rho_m)$ where α is the overlap on the same atom and γ is the overlap between the nearest neighbors connected by ρ_m and 'm' runs over all the nearest neighbors. (a) Now, derive ε_k for simple cubic structure considering its nearest neighbors. (b) Calculate the effective mass tensor M_{mn} at Γ point (k=(0,0,0)), at the face center/X-point (k=(0,1,1)) and at the corner/L-point (k=(1,1,1)). Here,tensor indices m and n runs over x,y and z directions. (2+3)

20. Using central equation, derive the Kroning-Penny Model expression $\frac{mAa^2}{2\hbar^2}\frac{sin(\alpha a)}{\alpha a} + cos(\alpha a) = cos(ka)$ using a periodic delta function potential, $U(x) = 2\sum_{G>0} U_G cos(Gx) = As\sum_s \delta(x - sa)$ where A is a constant and a is the lattice spacing. Here the boundary conditions are periodic over a ring of unit length, meaning over 1/a atoms. Here, $\alpha = \frac{2m\epsilon}{\hbar^2}$ with ϵ being the electron energy. (Hint: Kronig-Penny Model in Reciprocal space and $ctn(x) = \sum_n \frac{1}{n*\pi+x}$) (3+1+1)

21. (a) Assume that the real part of the conductivity of a superconductor is described by Re $\sigma(\omega) = A\delta(\omega)$ with $\delta(\Omega)$, being the Dirac delta function. This expression can be taken as the $\tau \to \infty$ limit of the real part of AC Drude conductivity, $\sigma(\omega) = \frac{\sigma_0}{1-i\omega\tau}$ with $\sigma_0 = \frac{ne^2\tau}{m_e}$ where e is the electron's charge, τ is the scattering time and m_e is the electron's mass.
(a.1) Express A in terms of the electron density, mass and charge.
(a.2) Evaluate the integral $\int_{-\infty}^{+\infty} d\omega \, \mathrm{Re}\, \sigma(\omega)$ (1.5+1)
(b) The constant temperature gradient $\frac{dT}{dX} = -\gamma < 0$ exists in a metal bar whose left end is maintained at temperature T_h and is taken to be the origin of the x axis.
(b.1) Compute the thermal energy difference per electron between $x - vt$ and $x + vt$ where v is the velocity of the electron.
(b.2) Evaluate the thermal current density at x (Assume a electron density of 'n' and note that heat/thermal energy is proportional to the temperature of the electron)
(1.5+1)

Susip P Sarai
02/12/23

EndSem Exam
PH4102: Introductory Astrophysics

Answer Question 1 and any 4 out of the 5 remaining questions.
Note: Meaning of a variable is same as we used in the class, unless stated otherwise.

Time : 150 min Marks: 50

(Given : $m_e = 9.1 \times 10^{-31}$ kg, $m_H = 1.67 \times 10^{-27}$ kg, $h = 6.62 \times 10^{-34}$ m^2 kg/s, $k_B = 1.38 \times 10^{-23}$ J/K, $G = 6.67 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, 1 AU $= 1.5 \times 10^8$ km)

1. a) What is the distance (in parsecs) of a star whose absolute magnitude is +6.0 and apparent magnitude is +16.0? (2)

b) Near the orbit of the Earth, the solar wind has a velocity of about 400 km/s and contains about 10 protons per cm^3 . Assuming that the solar wind always had these characteristics during the Sun's lifetime of 4.5×10^9 yr, estimate the fraction of mass the Sun would have lost in the solar wind during its lifetime. (2)

c) Virial theorem for stellar dynamics states that

$$2 < K > + < V >= 0,$$

<> represents time averaged quantities, and K and V are total kinetic and gravitational potential energies, respectively. Show, how astronomers use this theorem for computing the total mass of a star cluster, located at a certain known distance. (2)

d) How is the displacement current term neglected in MHD? (2)

e) If (r, ϕ) represent the polar coordinates, the differential distance between two points in flat space can be written by

$$dl^2 = (dr)^2 + (rd\phi)^2.$$

Show that this differential distance on a sphere's surface is given by

$$dl^2 = (\frac{dr}{\sqrt{1 - Kr^2}})^2 + (rd\phi)^2,$$

where, K is the curvature of the two-dimensional spherical surface. (2)

2. Consider a model star in which the density $\rho(r)$ goes as ρ_0 at the core ($r < r_0$), $\rho_0(r_0/r)^2$ in the region between the core and surface ($r_0 < r < R$) and 0 outside the surface ($r > R$).
a) Find an expression for $M(r)$.

b) If the star's mass is $1M_\odot(=2\times10^{30}$ kg) at $R = R_\odot(= 7\times10^5$ km) and $r_0 = 0.1R_\odot$, what is the value of ρ_0? (6+4)

3. a) Dense cores of giant molecular clouds (GMC) have following characteristic : Temperature $\approx$ 10 K, number density of hydrogen molecule (H_2) $\approx 10^{10}$ /m^3 and characteristic mass $\approx 10M_\odot$. Show that such dense cores are unstable to gravitational collapse (i.e., possible locations for star formation!). Derive any necessary formula that you use and state the assumptions clearly.

b) Now, for a typical diffuse hydrogen cloud, temperature $\approx$ 50 K, number density of hydrogen atom (H) $\approx 5\times10^8$ /m^3 and characteristic mass $\approx 100M_\odot$. Show that these clouds are stable to gravitational collapse (i.e., don't collapse).

(7+3)

4. a) Derive Euler equation for fluid dynamics.

b) How does this equation get modified in the case of magneto-hydrodynamics (MHD)? Explain the physical significance of additional terms. (5+5)

5. Inside a galaxy, the distribution of stars is believed not to be due to collisional relaxation. Prove this fact by showing that the collisional relaxation time for galaxy is several order larger than the age of the Universe. Assume typical mass of stars in the galaxy to be $1M_\odot$, their typical velocity 100 km/sec and typical number density 0.1 per parsec^{-3}. (10)

6. a) The reason for the white dwarf to be hotter even though they don't have any nuclear burning, is that their cooling time scale is very high. Make a rough estimate of white dwarf cooling time scale assuming the following:
i) Mass of white dwarf is $0.8M_\odot$ and made up of ^{12}C particles.
ii) Average internal temperature 10^7 K.
iii) White dwarf has a constant luminosity of 3.84×10^{23} J/s.

b) Consider stars of mass $1M_\odot$. Compute the mean mass density for the following:
i) Sun ($R_\odot = 7\times10^5$ km)
ii) a white dwarf ($R = 10^4$ km)
iii) a neutron star ($R = 10$ km)
Now consider a ^{12}C nucleus of radius $r = 3\times10^{-15}$ m and compute its mean density. Discuss the significance of all these results! (assume mass of proton and neutron to be same). (3+7)

Susip P Sarai
02/12/23

PH4102 Introductory Astrophysics
End Semester Supplementary 2022-2023
Saturday, 2 December 2023

Total Marks: 50 | Time: 2.5 Hours

Useful Constants

$M_\odot = 2 \times 10^{30}$ kg; $R_\odot = 7.0 \times 10^8$ m; $L_\odot = 3.826 \times 10^{26}$ J/s; Solar mean density = 1500 kg m^{-3}; Density of the solar core = 1.6×10^5 kg m^{-3}; Solar core temperature = 1.5×10^7 K; Solar surface temperature = 5,800 K; Age of the Sun = 5×10^9 years; 1 AU = 1.5×10^{11}m; Earth's radius = 6,400 km; $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}; Earth's gravity, g = 9.81 ms^{-2}; $m_e = 9.11 \times 10^{-31}$ kg ; Mass of hydrogen atom, $m_H = 1.67 \times 10^{-27}$ kg; 1 amu = 1.66×10^{-27} kg ; Mass of helium nucleus = 4.002603 amu; Mass of carbon nucleus = 12.00 amu; $K_B = 1.38 \times 10^{-23}$ J/K; $\sigma_{SB} = 5.67 \times 10^{-8}$ Wm^{-2}K^{-4}; Wien's constant = 2.89×10^{-3} m K; $h = 6.62 \times 10^{-34}$ m^2 kg s^{-1}; Wavelength of visible light = 550 nm; 1 erg = 10^{-7} joule; 1 J = 6.242×10^{18} eV; $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}; Speed of light in vacuum = 3×10^8 m s^{-1}

Use your own digital scientific calculator and answer all questions. Make any necessary assumptions required to solve any of the questions, however, state them clearly in your answer scripts. No further clarifications will be provided during the exam.

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1. (a) The Sun rotates on its axis once in about 27 days – state a simple observational evidence in favour of this rotational motion.

   (b) How does the Sun's rotation introduce spectral shifts in its emission spectrum across the solar disk?

   (c) Spatially resolved stars (e.g. the Sun) are observed to be darker at the edge i.e. the limb; than at the centre of the disk – explain in details.

   (1 + 1 + 3 = 5)

2. (a) Can we expect electrons in a neutron star? Explain!

   (b) If the star Proxima Centauri ($M = 0.123 M_{Sun}$) collapses to a compact object such that escape velocity at its surface exceeds the speed of light, what would be its maximum radius?

   (2 + 3 = 5)

3. Suppose that you are attempting to make observations through an optically thick gas that has a constant density and temperature. Assume that the density and temperature of the gas are $2 \times 10^{-4}\, kg/m^3$ and $6000K$, respectively, typical of the values found in the Sun's Photosphere.

(a) If the opacity of the gas at one wavelength (λ_1) is $\kappa(\lambda_1) = 0.021\, m^2/kg$ and the opacity at another wavelength (λ_2) is $\kappa(\lambda_2) = 0.039\, m^2/kg$, calculate the distance into the gas where the optical depth equals 2/3 for each wavelength. Hint: Use the relationship between optical depth and opacity.

(b) At which wavelength can you see farther into the gas and how much farther?

(c) Can this effect help us to probe various solar atmospheric depths? Explain!

$(2 + 1 + 2 = 5)$

4. (a) Suppose a star has an inverse square density profile. The particles inside the star satisfy the condition $p << mc$, where p is the total momentum and c is the speed of light. What is the significance of calculated energy?

(b) What are the possible origins of pressure inside a massive star.

(c) For the stars dominated by the relation $p >> mc$, calculate the total energy. Interpret your result.

(d) Establish the scaling relationship between gas pressure, radiation pressure and mass. What can you conclude from this relationship?

Hint: Assume quantum mechanical effects are unimportant.

$(2 + 2 + 2 + 2 = 8)$

5. (a) Assume a gravitationally bound star of mass M which follows a constant density profile. Use the virial theorem to show that

$$P \sim (\frac{4\pi}{3^4})^{1/3} G M^{2/3} \rho^{4/3},$$

(b) Show that if the radiation pressure equals the kinetic pressure, then the total pressure is

$$P = 2\left(\frac{3}{a}\right)^{1/3}\left(\frac{\kappa\rho}{\bar{m}}\right)^{4/3}.$$

(c) Obtain an expression for the maximal mass of a stable star. Find its approximate value in Solar masses, assuming a fully ionized hydrogen composition.

(where $\bar{m}$ is the average mass per particle, κ = Boltzman constant, a = $4\sigma/c$, where σ is Stefan-Boltzmann's constant.)

(3 + 3 + 4 = 10)

6. (a) Assume a hypothetical scenario in which the temperature-radius dependence of a star is inverted. For such a strange star, what would you expect the radiation spectrum of the star to look like for those frequencies at which the absorption coefficient is highest? Justify your answer in your own words.

   (b) Consider sunlight streaming into your room through a window. Explain using logic whether you expect the air in the room to be in equilibrium with the radiation.

(3 + 3 = 6)

7. Consider a white dwarf formed due to contraction of thermally unsupported stellar core of mass $M = M_\odot$ down to radius R_w at which degeneracy pressure inhibits further contraction. Assuming $\frac{Z}{A} = 0.5$ where Z and A are atomic number and mass number respectively,

   a) Find a mathematical expression for R_w in terms of Z, A, and M (Assume that the stellar core consists of ionized helium only. Use scaling approximations of the relevant equations).

   b) Find the temperature of the degenerate core of mass $0.5M_\odot$ for pure helium composition.

   c) Since white dwarfs are end-points in stellar evolution devoid of nuclear reactions, a white dwarf, therefore, cools down by radiating from its surface the thermal energy stored in the still-classical gas of nuclei within the star's volume. Find an expression of cooling time and show how it varies with temperature. (Assume a constant temperature all the way out to the surface of the star).

(3 + 3 + 2 = 8)

8. The HR Diagram denotes the distribution of various stars in a plot with luminosity in the y-axis and temperature in the x-axis (decreasing temperatures to the right).

   (a) Plot these axes, and indicate the Main Sequence track. Where do you expect very high mass (but normal) Main Sequence stars to be located? Explicitly mark this location with a "O" symbol.

(b) Where do you expect newly born Brown Dwarf stars to be located in this diagram? Explicitly denote the location of young Brown Dwarfs on your diagram with a "X" symbol.

$(2 + 1 = 3)$

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Nonlinear Dynamics (PH4104): End Semester Examination

Instructor: Dr. Rumi De; Date: 30.11.2023; Total marks: 50

1(a) Consider a two-dimensional dynamical equation, $\dot{x} = -mx$, $\dot{y} = -y$; Plot the phase portrait for **(i) m=1** and **(ii) m=0** and justify your answer. **1+1=2**

(b) Consider a damped nonlinear oscillator, $\ddot{x} + \mu\dot{x} + x - x^3 = 0$; verify whether the system has any limit cycle solution. **2**

(c) Calculate the Lyapunov exponent for the linear map, $x_{n+1} = rx_n$. **2**

(d) What is a multifractal system? Explain it with an example. **2**

2. Consider the dynamical system, $\dot{y} = my - \ln(1+y)$, with parameter m.
(a) Sketch all the qualitatively different vector fields that occur as **m** is varied
(b) Finally, sketch the bifurcation diagram of fixed points versus **m**. Also, identify the type of bifurcation. **1.5+1.5=3**

3. Consider the map $x_{n+1} = 1 + \frac{1}{x_n}$.
(a) Find out the fixed points and analyze their stability.
(b) Draw cobwebs for two different initial conditions: **(i)** $x_0 = -1$ and **(ii)** $x_0 = 2$. **1+2=3**

4. To construct the famous fractal "Koch snowflake" curve, consider an equilateral triangle for S0. Then, to generate S1, follow the Koch curve procedure on each of the three sides of the triangle.
(a) Draw S0, S1, and S2.
(b) Find the fractal dimension of the set. **1.5+1.5=3**

5. Consider the dynamical equations, $\dot{x} = x + 2xy$, $\dot{y} = x + x^2 - y^2$;
(a) Find all the fixed points and analyze the stability of the fixed points.
(b) Show whether the system has any limit cycle or not.
(c) Finally, evaluate the flow directions and draw the complete phase portrait. **1+1+2=4**

6. Glycolysis is a fundamental biochemical process in which living cells obtain energy by breaking down sugar. The glycolysis process can be modeled by the following dynamical equations:

$$\dot{x} = -x + ay + x^2y,$$
$$\dot{y} = b - ay - x^2y;$$

Where x and y are the concentrations of ADP and F6P, and a, b >0 are kinetic parameters.
(a) Sketch the nullclines and (b) construct a trapping region for this system. **2+3=5**

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7. Consider a two-dimensional dynamical system,
$\dot{x} = \mu x - \omega y + K\,x\,(x^2 + y^2)$, $\dot{y} = \omega x + \mu y + K\,y\,(x^2 + y^2)$; with parameters μ, ω, and K.
We consider μ as the varying parameter and $\omega > 0$, and $K > 0$.
(a) Find the fixed points of this system and analyze the stability of the fixed points as the parameter μ is varied.
(b) Use the Poincare-Bendixson theorem to show that a limit cycle exists in this system for a specific range of parameter values.
(c) Analyze the stability of the limit cycle and draw the phase portraits for different parameter values. Also, illustrate it in a bifurcation-like diagram. **1+2+3=6**

8. Consider the cubic map, $x_{n+1} = r x_n - x_n^{\,3}$,
(a) Find all the fixed points and analyze their stability.
(b) Suppose p, q be the two cycles of the map. Show that p, q are roots of the equation,

$$x(x^2 - r + 1)(x^2 - r - 1)(x^4 - rx^2 + 1) = 0.$$

(c) Determine the stability of the two cycles as a function of r. **1+2.5+2.5=6**

9. Dynamics of the budworm population can be described by the following equation,

$$\dot{x} = rx\left(1 - \frac{x}{k}\right) - \frac{x^2}{1+x^2}, \text{ with parameters } \mathbf{r} \text{ and } \mathbf{k}.$$

(a) Show that the fixed point **x*=0** is always unstable.
(b) Analyze the stability of all the fixed points by graphical representations and sketch qualitatively different vector fields that occur as **r** and **k** are varied.
(c) Determine the limiting behaviour of **r(x)** and **k(x)** as $x \to 1$ and $x \to \infty$. Sketch a qualitative bifurcation diagram **r(x) vs x** (for fixed k) and show that there is a possibility of a budworm outbreak.

1+2+3=6

10. Brusselator model describes the spatiotemporal evolution of two chemical concentrations of u_1 and u_2 following the given reaction-diffusion equations:

$$\frac{\partial u_1}{\partial t} = a - (b+1)u_1 + u_1^2 u_2 + D_1 \frac{\partial^2 u_1}{\partial x^2}$$

$$\frac{\partial u_2}{\partial t} = b u_1 - u_1^2 u_2 + D_2 \frac{\partial^2 u_2}{\partial x^2};$$

where D_1 and D_2 are the diffusion coefficients, and **a, b** are the parameters.
(a) Considering a stable homogeneous steady state, derive and analyze the conditions for Turing instability for this system.
(b) Find the condition for the onset of instability and the spacing between chemical concentration peaks. **3+(2+1)=6**

PH4106 (Autumn - 2023)
Basics of Field Theory and Relativistic Quantum Mechanics
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

End-Semester Examination Time: 2.5 Hrs

1. (**Total Marks: 11**) The Dirac Lagrangian is given by (**Total Marks: 11**)

$$\mathcal{L}_D = \bar{\psi}(i\partial\!\!\!/ - m)\psi \ . \tag{1}$$

(a) Find out equation of motions for ψ and $\bar{\psi}$ field, and show that those equations of motions are conjugate to each other. **Marks: 2+2**

(b) Find out the conserved charge associated with a continous internal symmetry of the Lagrangian. **Marks: 3**

(c) Show that each component of the Dirac spinor ψ satisfies the Klein Gordon equation. **Marks: 4**

2. (**Total Marks: 4**) Write down the Dirac equation in the following form:

$$\mathcal{L}_D = \bar{\Phi}\mathcal{O}\Phi + \text{total derivatives} \ , \tag{2}$$

where $\bar{\Phi}$ is conjugate of Φ such that $\bar{\Phi}\Phi$ is a Lorentz scalar. Using the form of $\mathcal{O}$, find the expression for the Feynman propagator $i\Delta_F(p)$ for a Dirac spinor. Show steps clearly.

3. (**Total Marks: 5**) In the class, we defined $\sigma^{\alpha\beta} = \frac{i}{2}[\gamma^\alpha, \gamma^\beta]$, where γ^αs are the usual Dirac matrices. Show that $(\sigma^{ij})^\dagger = \sigma^{ij}$, and $(\sigma^{0i})^\dagger = -\sigma^{0i}$.

4. (**Total Marks: 10**) Consider the following interaction Hamiltonian density $\mathcal{H}_{int} = gC(A^2 + B^2)$ where A, B and C are three scalar fields with A and B having equal mass m, and C has the mass M.

(a) For $A+B \to A+B$, write down the Feynman diagram(s) and the associated Feynman amplitude in terms of external momenta. **Marks: 4**

(b) For $A + A \to A + A$, write down the Feynman diagram(s) and the associated amplitudes in terms of external momenta. **Marks: 6**

5. (**Total Marks: 20**) Consider a part of the Quantum Electrodynamics (QED) Lagrangian given by

$$\mathcal{L} = \bar{\psi}_e(i\partial\!\!\!/ - m)\psi_e - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + e\bar{\psi}_e\gamma^\mu\psi_e A_\mu \ , \tag{3}$$

where ψ_e corresponds to the electron (e) field and A_μ is the photon (γ) field.

(a) Write down the mode decomposition of ψ_e, $\bar{\psi}_e$, and A_μ field operators. **Marks: 2**

Consider the following process (Compton scattering) $e(p,s) + \gamma(k,r) \to e(p',s') + \gamma(k',r')$.

(b) Write down the Feynman rules for the incoming and outgoing electrons and photons. **Marks: 4**

(c) Draw relevant Feynman diagram(s) for the above process at tree-level and mark each part of the diagram properly. **Marks: 6**

(d) Write down the total Feynman amplitude for the process. **Marks: 6**

(e) From the Dyson series expansion, identify the term that gives the above diagrams. **Marks: 2**

6. **(Total Marks: 18)** Consider the following classical Lagrangian:

$$\mathcal{L}(\psi, \psi^*) = i\psi^* \frac{\partial \psi}{\partial t} + \lambda \vec{\nabla}\psi^* \cdot \vec{\nabla}\psi \ , \tag{4}$$

where λ is a real number.

(a) Find out the Euler-Lagrange equations for the fields ψ and ψ^*. They should be complex conjugate of each other. **Marks: 3**

(b) For a plane wave solution, find out angular frequency ω in terms of $\vec{p}$. **Marks: 1**

(c) The above Lagrangian is space-time invariant (you do not need to show). Find out the component of the energy momentum tensor T^{00}, and derive an expression for the total Hamiltonian H. What condition do you get on λ by demanding that Hamiltonian is positive definite. **Marks: 3**

(d) Chose λ in such a way that the equation of motion in part (a) looks like the Schrödinger equation for a nonrelativistic particle of mass m. How does ω looks in part (b) now? **Marks: 2**

(e) Show that the total Hamiltonian can be written as expectation value of the kinetic energy of a free particle. **Marks: 3**

(f) Find out T^{i0}, and show that it equivalent to the expectation value of the momentum operator. **Marks: 2**

(g) The Lagrangian also has an internal symmetry. Write down the symmetry, and find out the charge associated with the symmetry. How do you interpret the charge? **Marks: 2**

(h) Does the system has Lorentz symmetry? **Marks: 1**

(i) In maximum two sentences, write a few words (your understanding) about the system described by the above Lagrangian. **Marks: 1**

7. **(Total Marks: 12)** Consider the following Lagrangian for two real Klein-Gordon field:

$$\mathcal{L}_{\text{fundamental}} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 + \frac{1}{2}\partial_\mu\chi\partial^\mu\chi - \frac{1}{2}M^2\chi^2 - \frac{1}{2}g\phi^2\chi \ . \tag{5}$$

We want to study $\phi(p_1) + \phi(p_2) \to \phi(p_3) + \phi(p_4)$ scattering process.

(a) Draw all Feynman diagrams(s) for the above mentioned process and write down the total Feynman amplitude in terms of $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$ and $u = (p_1 - p_4)^2$. Mention clearly your conventions, vertex factor and expression for the propagator. **Marks: 5**

(b) Now, we are interested in the case of $m \ll M \sim g$, i.e. the χ particle being much heavier than the ϕ particle. Suppose, a low energy observer sets out to study the above mentioned scattering at a center-of-mass energy $E = \sqrt{s} \ll M$. In this case, typically $t \ll M$ and $u \ll M$ also. Expand the Feynman amplitude in part (a) in inverse powers of M upto $\mathcal{O}(1/(\text{mass})^4)$, and arrange terms in inverse powers of M. **Marks: 3**

(c) A low-energy observer will see only the dominant lowest order term in (b). What would be the effective Lagrangian $\mathcal{L}_{\text{effective}}$ for this low energy observer to describe $\phi(p_1) + \phi(p_2) \to \phi(p_3) + \phi(p_4)$ scattering process? Identify the parameter in $\mathcal{L}_{\text{effective}}$ in terms of the parameters in $\mathcal{L}_{\text{fundamental}}$. **Marks: 4**

This is roughly the end of the course! But, surely not your learning ..
Do good :-)

IISER Kolkata
Department of Physical Sciences
Advanced Electricity, Magnetism and Optics, PH4107
Autumn Semester 2023
Final Examination, dated 01/12/2023
Max. Marks: 40

Attempt any four

Q1. 4+3+3

(a) Find out the 1×4 Stokes vectors corresponding to the following 1×2 Jones vectors
(i) $2 * [0.4 \quad 0.3]^T$ (ii) $[0.6 \quad 0.8\, e^{i\pi/3}]^T$ (iii) $\left[1 \quad e^{-\frac{i\pi}{2}}\right]^T/\sqrt{2}$
Comment on the polarization state and the degree of polarization of each of the states.

(b) Consider the following results of polarized light experiment, where input linearly polarized light (with Stokes vector $[1 \quad 1 \quad 0 \quad 0]^T$) is backscattered from two different depolarizing samples. The measured Stokes vectors from the two samples are - $[1 \quad 0.6 \quad 0.3 \quad 0.2]^T$ and $[1 \quad 0.4 \quad 0.4 \quad 0.7]^T$ respectively. Which of the two samples depolarize light more? What are the polarized fractions of the total intensity in the two samples?

(c) Show that two cascaded quarter-wave retarders with parallel fast axes orientation are equivalent to a half-wave retarder. What is the result if the fast axes are orthogonal?

Q2. 3+4+3

(a) In which one (s) among the following three cases, can the light scattering be described using the quasi-static approximation (QSA)?
(i) Diameter of scatterer d = 30 nm, refractive indices of the scatterer and the surrounding medium, $n_s = 5$ $n_m = 1.33$ respectively, wavelength λ =600 nm. (ii) d= 10 nm, n_s =1.38, n_m = 1.33, λ =600 nm. (iii) d = 400 nm, n_s =1.38, n_m = 1.33, λ =600 nm.
In which case would the first order Born approximation be valid but not the QSA? Explain your answer based on the criteria for validity of the approximations.

(b) The absorption efficiency of a small metal sphere with dielectric permittivity ϵ ($=\epsilon' + i\epsilon''$), embedded in a medium with dielectric permittivity ϵ_m, is given by

$$Q_{abs} = 4x\, Im\left\{\frac{\epsilon - \epsilon_m}{\epsilon + 2\epsilon_m}\right\}$$

Here, x is the size parameter of the scatterer. Assuming ϵ_m to be real, show that the absorption efficiency of the scatterer at the Frohlich frequency ω_F (corresponding to dipolar resonance) will vary as $Q_{abs} \propto \frac{1}{\epsilon''(\omega_F)}$. Can you explain this counterintuitive behavior: maximum absorption is inversely proportional to the absorptive part of the complex dielectric permittivity?

(c) Two polarizers can be used as a continuously variable grey filter. What is the angle between the transmission axes, so that 5.0 % of incoming light is transmitted?

Q3. 5+5

Consider a spherical void (permittivity $\epsilon = 1$) with radius a = 10 nm embedded in an otherwise homogeneous metallic medium with permittivity $\epsilon_m(\omega)$. The wavelength λ_p corresponding to the plasma frequency of the metal is given to be 250 nm.

(a) Determine the wavelengths corresponding to the dipolar (λ_d) and quadrupolar (λ_q) plasmon resonances of the void.

(b) Compare the dipolar and quadrupolar plasmon resonance wavelengths of the spherical void with that of a metal sphere (a = 10 nm, permittivity $\epsilon_m(\omega)$) embedded in air ($\epsilon = 1$). In which case (void or sphere) would the plasmon resonance wavelengths be higher? Briefly explain the physical basis of the differences in the resonance wavelengths.

[Use Drude's ideal free electron model for dielectric permittivity of metal. Use the condition of resonances for the dipolar and the quadrupolar polarizabilities.]

Q4. 3+4+3

The complex transparency distribution of an object is given by the following function

$$t(x) = A\left[\frac{x}{(x^2+\alpha^2)} - i\frac{\alpha}{(x^2+\alpha^2)}\right], \text{ where } \alpha \text{ is a spatial scaling parameter}$$

(a) Plot the amplitude and the phase distribution at the object plane around $x = 0$.

(b) Derive expression for the intensity of Fraunhofer (far field) diffracted waves from the object as a function of the spatial frequency $\left(\nu_x = \frac{x_i}{\lambda r}\right)$ of the object. Here, x_i is the spatial co-ordinate at the Fourier (diffraction) plane and r is the distance from the object centre. Sketch the corresponding spatial frequency distributions at the Fourier plane.

(c) Draw an analogy of the above with the causal temporal response function of an optical system.

Q5. 4+2+4

(a) Consider scattering of light by a dielectric sphere of radius a having refractive index (RI) n_s embedded inside a medium with RI n_m [a <<λ, (n_s / n_m - 1) << 1]

The differential scattering cross-section for incident X-polarized light is

$$\frac{d\sigma(\theta,\phi)}{d\Omega} = K[(cos^2\theta + 1) + (cos^2\theta - 1)cos2\phi]$$

Where, ϕ is the azimuthal angle measured from the direction of linear polarization (X), θ is the scattering angle and K depends upon the size and RI of the scatterer. What would be the corresponding expressions for incident left circularly, right circularly and unpolarized light?

(b)"Measurements of the scattered field in the far zone can at best provide information about the structure of the scatterer down to a resolution of the order of half a wavelength". Is the statement correct? Justify your answer.

(c) The scattered intensity (I_s) of light from a scattering object is found to vary as scattering angle (θ)as $I_s(\theta) \approx (1 + cos^2\theta)$. Estimate the scattering anisotropy parameter (defined as the average cosine of the scattering angle) of the object.

Indian Institute of Science Education and Research, Kolkata
Department of Physical Sciences
Biological Physics (PH4108)
Instructor: *P. K. Mohanty*

28/11/2023

Endsem Exam (28th Nov, 2023, 10:00-12:30) **Full Mark: 30**

Attempt any 5 *questions out of* 8. *Each carry* [6] *marks.*

Notations: k_B is Boltzman constant. T is temperature. Noise $\eta(t)$ has PDF $\mathcal{N}(0,1)$. $\delta(.)$ is the Dirac delta function. $C_l^m = \frac{m!}{l!(m-l)!}$ is the Binomial function. $\sum_{i=0}^{n} i^2 = \frac{1}{6}n(n+1(2n+1)$.

1. HSP70 is a major stress-induced heat shock protein that repairs and protects cells after insult. Prolonged presence of this protein is however detrimental. Expression of HSP70 is tightly regulated: the DNA of the cell under stress transcribe (HSP70)-mRNA at rate k which is then translated to proteins at rate p. These short-lived proteins degrade at rate d. Assume concentration of (HSP70)-mRNA and HSP70 protein at time t are respectively $m(t), x(t)$ with initial conditions $m(0) = 0, x(0) = 0$. Show that $m(t) = \frac{k}{p}(1 - e^{-pt})$. Use this in the evolution equation of $x(t)$ and solve it to show that in long time limit HSP70 is produced at a constant rate $(k - d)$ (independent of p !). [6]

2. (a) For any integer $n \geq 0$, functions $F_n(x) = \frac{x^n}{n!}e^{-x}$ are normalized when $x \in (0, \infty)$. Show that PDF of $z = x + y$ is $F_{k+m+1}(z)$, if PDF of $x \geq 0$ is $F_k(x)$ and PDF of $y \geq 0$ is $F_m(y)$. **Hint:** You may use the Identity $\sum_{l=0}^{m} \frac{(-1)^l}{k+l+1} C_l^m = \frac{k!m!}{(k+m+1)!}$. [3]
   (b) Draw three independent positive random numbers (x_1, x_2, x_3) from an exponential distribution e^{-x} (normalized for $x \in (0, \infty)$. What is the PDF of $z = x_1 + x_2 + x_3$? [3]

3. Enzymatic reactions happen as follows. An enzyme E interacts with a substrate S to produce a complex ES, which then degrades to a product P returning the enzyme to its original state.

$$E + S \underset{2r}{\overset{r}{\rightleftharpoons}} ES \xrightarrow{2r} E + P.$$

   Since Enzymes remain as they are, one can describe the reaction kinetics as transitions among these three states "S", "ES" and "P." with densities $(x(t), y(t), z(t)) \equiv |P\rangle$. If substate S is supplied to the system with rate α, at what rate product P will be formed in a large time limit ? **Hint:** Write $\frac{d}{dt}|P\rangle = M|P\rangle + |v\rangle$, use $M = UDU^{-1}$ and rewrite the Master equation in terms of $|\tilde{P}\rangle = U^{-1}|P\rangle$ and solve for $z(t)$ only. [6]

4. (a) Micro-RNAs (miRNAs) are small RNAs (single stranded) composed of bases A,U,G,C, where A can form a pairing with U and G can form a pairing with C. Any miRNA of $2N$ base long can form a loop if 1st base forms a pair with $2N$, N-th base pairs with $N+1$, and all other bases remain unpaired. How many loops can a 12-base miRNA form ? [3]

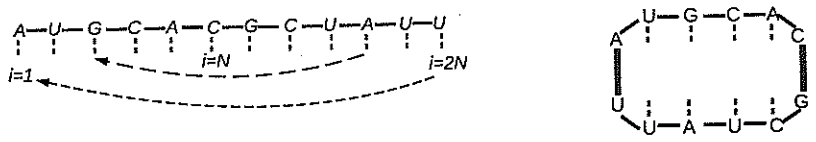

Figure 1: miRNA (left) of length $2N$ folds to form a loop (right) by joining k-th site with (2N+1-k)-th site $k = 1, 2, N$

   (b)How many miRNAs has no complementary pairing at any of the 12 sites ? [2]
   (c) How many miRNAs are perfectly-bonded in folded condition ?[1]

5. On a periodic lattice of size $L = 3$, Ising spins $s_i = \pm 1$ are placed at $i = 1, 2, 3$ such that the total magnetization $M = \sum_{i=1}^{3} s_i$ is fixed at $M = 1$. Energy of the system is given by

$$E(\{s_i\}) = -\sum_{i=1}^{3} h_i s_i + U s_i s_{i+1} s_{i+2},$$

where $h_i = 3i^2$ is an inhomogeneous magnetic field and U is the strength of three spin interaction. Obtain the Master equation that evolve the system following Metropolis rates

$$W_{C\to C'} = \text{Min}\{1, exp(-\beta(E(C') - E(C)))\}.$$

Show that the steady state probability of configurations $\{s_i\}$ of the system are given by the Boltzmann distribution, $P^*(\{s_i\}) = \frac{1}{Z}\, e^{-\beta E(\{s_i\})}$, where Z is a normalization factor. [6]

6. DNA is a double stranded sequence of two chains, consisting four bases (A,T,G,C) with pairing convention A-T bond with energy $-\alpha$ and G-C having bond energy $-\gamma$. The bonds that form the linear chains are stable and does not break at any temperature. See the figure below for a reference.

```
.....A—T—G—T—A—C.....
      |  |  |  |  |  |
.....T—A—C—A—T—G.....
```

Assuming that the energy of broken bond is 0, irrespective of whether it is A-T or G-C, find the probability that a DNA strand of length L with N A-T bonds denatures completely (all bonds broken) at temperature T ? [6]

7. In one dimension, thermal fluctuation of a molecule composed of two identical atoms can be modeled by a massless spring (equilibrium length l and spring constant k) with point masses m attached at both ends. Position of these atoms (masses) $x(t)$ and $y(t)$ fluctuates at temperature T as follows.

$$\begin{aligned} m\ddot{x} &= -k(x - y + l) - \gamma\dot{x} + \sqrt{2k_BT}\eta_x(t) \\ m\ddot{y} &= -k(y - x - l) - \gamma\dot{y} + \sqrt{2k_BT}\,\eta_y(t) \end{aligned}$$

(a) Show that the center of mass $z = \frac{x+y}{2}$ behaves like a free particle in a Langevin bath [3]
(b) What is the variance of the center of mass velocity $\dot{z}$. [2]
(c) In the overdamped limit, find the variance of z. [1]

8. The zero range process (ZRP) is defined on a one dimensional periodic lattice with sites (or boxes) labeled by $i = 1, 2, \ldots L$; box i containing n_i number of particles. Dynamics: particle from a randomly chosen box i hop to its right neighbor with a rate $u(n_i)$. The steady state probability of configuration $\{n_i\}$ are given by $P(\{n_i\} = \frac{1}{Q}\prod_{i=1}^{L} f(n_i)\,\delta(\sum_{i=1}^{L} n_i - N)$, where $f(n) = \prod_{k=1}^{n} \frac{1}{u(k)}$ and Q normalizes the probability. Let us consider two boxes ($L = 2$) containing N and particles from one box hops to the other box with rate $u(n) = \frac{n}{n+1}$. What is the probability that, in steady state, the first box carries α fraction of total particles ? Show that in a large N limit, the probability is proportional to N^{-1}; what is the proportionality constant ? [6]

PH 5103 end-sem Exam 01/12/2023

(Provide explicit details of calculations unless they are too obvious!)

1. Consider spin-$\frac{1}{2}$ electrons on a 2D square lattice of unit spacing described by:

$$\hat{\mathcal{H}} = t\sum_{\langle ij\rangle,\sigma}\left(\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma} + \text{H.c.}\right) - J\sum_{\langle ij\rangle}\left(\mathbf{S}_i\cdot\mathbf{S}_j - \frac{1}{4}n_i n_j\right) - \mu\sum_{i,\sigma}\hat{n}_{i\sigma}$$

Here, $\langle ij\rangle$, indicates that i and j are nearest neighbor lattice sites and $n_i = n_{i\uparrow} + n_{i\downarrow}$. Here, H.c. stands for Hermitian conjugate and t, μ and W are all real positive numbers. Other notations carry usual meaning.

(a) Expressing the spin operators in terms of electronic creation and annihilation operators, i.e. $\mathbf{S}_i = \sum_{\alpha,\beta}\hat{c}^{\dagger}_{i\alpha}\tau_{\alpha\beta}\hat{c}_{i\beta}$ (τ: Pauli matrices) and derive the corresponding Hamiltonian expressed in momentum space by employing Fourier transform of electronic creation and annihilation operators. Upon considering only the "reduced BCS interactions". Obtain the explicit expression for $V_{k,k'}$.

(b) Set up the mean field decomposition of the above Hamiltonian in terms of the non-zero order parameters: $\Delta_k = \sum_{k'} V_{k,k'}\langle\hat{c}_{-k\downarrow}\hat{c}_{k\uparrow}\rangle$ and $\rho_k = \sum_{k,\sigma}\langle\hat{c}^{\dagger}_{k\sigma}\hat{c}_{k\sigma}\rangle$ (assume $\rho_{k\uparrow} = \rho_{k\downarrow}$). Identify the form factors $g_p(k)$ for all possible p (in this present case), by writing $V_{k,k'} = \sum_p V_p g_p(k) g_p(k')$. Using the annsatz $\Delta_k = \sum_p \Delta_p g_p(k)$, and also with the assumption of $\Delta_k = \Delta_{-k}$, derive the *gap equation* (for Δ_p) and *density equation* for the given Hamiltonian.

[points: 14=6+8]

2. Consider a free classical electron in a magnetic field $\mathbf{B} = \nabla\times\mathbf{A}$. The gauge invariant momentum is written as $\mathbf{p} \to \mathbf{\Pi} = \mathbf{p} + e\mathbf{A}(\mathbf{r})$. The classical Hamiltonian is then $\mathcal{H} = (\mathbf{p} + e\mathbf{A}(\mathbf{r}))^2/2m$, where notations carry usual meaning, unless mentioned otherwise. Consider a 2D system on x-y plane and $\mathbf{B} = B\hat{z}$, and Show that the extension to quantum mechanics (similar to, $p_x \Rightarrow -i\hbar\partial/\partial x$ in the absence of magnetic field), leaves the gauge invariant momentum operator $\hat{\mathbf{\Pi}}$ non-commuting, i.e, show that: $\left[\hat{\Pi}_x, \hat{\Pi}_y\right] = -ie\hbar B$, unlike $[\hat{p}_x, \hat{p}_y] = 0$. Defining, $l_B^2 = \hbar/(eB)$, show

that, the ladder operators, defined as:

$$\hat{a} = \frac{l_B}{\hbar\sqrt{2}}\left(\hat{\Pi}_x - i\hat{\Pi}_y\right),$$

and similarly for $\hat{a}^\dagger$, satisfy the Bosonic commutation relation $\left[\hat{a}, \hat{a}^\dagger\right] = 1$. Show further that in terms of these ladder operators, the quantum version of the given Hamiltonian, which can be written as $\hat{\mathcal{H}} = \left(\hat{\Pi}_x^2 + \hat{\Pi}_y^2\right)/2m$, becomes $\hat{\mathcal{H}} = \hbar\omega_c\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$, where $\omega_c = \cancel{\hbar^2/ml_B}$. $\frac{\hbar}{m l_B^2}$

[points: 4+3+3=10]

3. In order to solve Shrödinger equation for an electron in 2D in the presence of a magnetic field $\mathbf{B} = B\hat{z}$, we used the Landau Gauge $\mathbf{A} = xB\hat{y}$ in class, resulting in the Hamiltonian: $\hat{\mathcal{H}} = \left(\hat{p}_x^2 + (\hat{p}_y + eB\hat{x})^2\right)/2m$. The corresponding Shrödinger equation was solved (in class) by separation of variable ansatz: $\psi_k(x,y) = e^{iky}f_k(x)$, using the translational invatiance in the y-direction. As a result, $\hat{\mathcal{H}}\psi_k(x,y) \equiv \hat{\mathcal{H}}_k\psi_k(x,y)$, where $\hat{\mathcal{H}}_k = \hat{p}_x^2/2m + \frac{1}{2}m\omega_c^2(\hat{x} + kl_B^2)^2$ is the hamiltonian of a shifted harmonic oscillator $\psi_{n,k}(x,y) \sim e^{iky}H_n(x + kl_B^2)$. Here, $H_n(x)$ are usual Hermite polynomial wavefunctions of the harmonic oscillator. Given these, find out the eigenfunctions $\psi_{n,k}(x,y)$ and eigen-energies upon switching on an electric field of strength E in the x-direction.

[points: 6]